Question 1

2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Every integer can be represented in one of six forms : $6k$, $6k+1$, $6k+2$, $6k+3$, $6k+4$ or $6k+5$. The numbers in the form $6k$, $6k+2$, $6k+4$ are even, while those in the form $6k+3$ are divisible by 3.

Therefore, these forms cannot produce prime numbers greater than 3. The only remaining possibilities are $6k+1$ and $6k+5$. Since $6k+5 = 6(k+1)-1$, every prime number greater than 3 must be expressible as $6k+1$ or $6k-1$, for some integer k.

CS Scanné avec CamScanner